Page 1: Environmental Corrosion Standards

Overview of atmospheric oxidation and corrosion prevention in fluid transmission lines and structural steel.

Page 2: Rust & Oxidation Treatment

Defect Type: Surface Corrosion and Rust.

Symptom: Brownish oxide flakes, surface pitting, or paint degradation on metallic piping.

Action Required: Scrape loose rust off using wire brush or sandblasting. Apply chemical rust converter followed by two coats of zinc-rich primer paint. Inspect wall thickness with ultrasonic gauge if pitting exceeds 1mm depth. Priority: Medium.

Page 3: Pitting and Wall Thickness Loss

Defect Type: Corrosion with wall thinning.

Symptom: Deep pits, or ultrasonic readings showing wall loss above 20 percent of nominal thickness.

Action Required: Take the line out of pressurised service. Cut out and replace the affected pipe section when wall loss exceeds 20 percent; between 10 and 20 percent schedule repair within 7 days and re-measure monthly. Priority: High.

Page 4: Protective Coatings and Cathodic Protection

Procedure: Preventing corrosion returning.

Re-coat repaired steel within 4 hours of surface preparation. Check sacrificial anodes every 6 months and replace any that have lost more than half their mass. Record coating thickness readings after each paint job.